Bangladesh Army University of Engineering & Technology
Department of Electrical & Electronic Engineering

Lecture: 10 - 11

Course Code: EEE 1261

Course Title: Electronic Devices and Circuits

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Designation: Lecturer, Dept. of EEE, BAUET

2.7 FULL-WAVE RECTIFICATION

Bridge Network

The dc level obtained from a sinusoidal input can be improved 100% using a process called *full-wave rectification*. The most familiar network for performing such a function appears in Fig. 2.53 with its four diodes in a *bridge* configuration. During the period $t = 0$ to $T/2$ the polarity of the input is as shown in Fig. 2.54. The resulting polarities across the ideal diodes are also shown in Fig. 2.54 to reveal that D_2 and D_3 are conducting, whereas D_1 and D_4 are in the "off" state. The net result is the configuration of Fig. 2.55, with its indicated current and polarity across R. Since the diodes are ideal, the load voltage is $v_o = v_i$, as shown in the same figure.

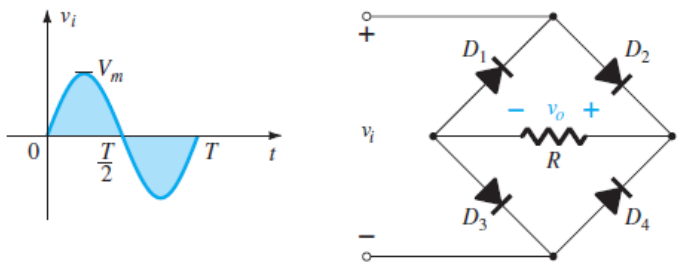

FIG. 2.53
Full-wave bridge rectifier.

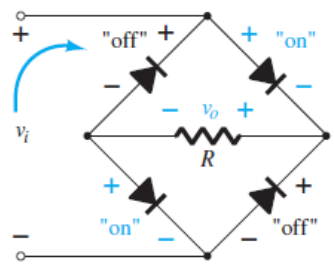

FIG. 2.54
Network of Fig. 2.53 for the period $0 \rightarrow T/2$ of the input voltage v_i.

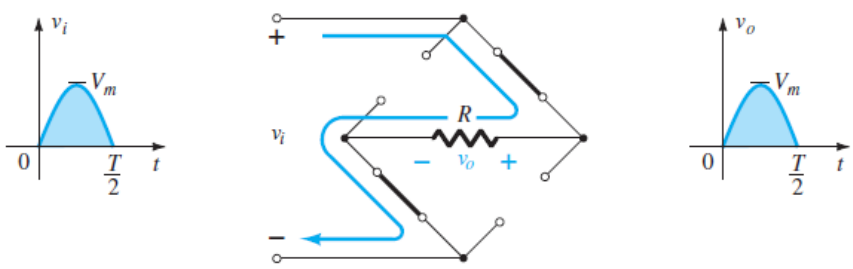

FIG. 2.55
Conduction path for the positive region of v_i.

For the negative region of the input the conducting diodes are D_1 and D_4, resulting in the configuration of Fig. 2.56. The important result is that the polarity across the load resistor R is the same as in Fig. 2.54, establishing a second positive pulse, as shown in Fig. 2.56. Over one full cycle the input and output voltages will appear as shown in Fig. 2.57.

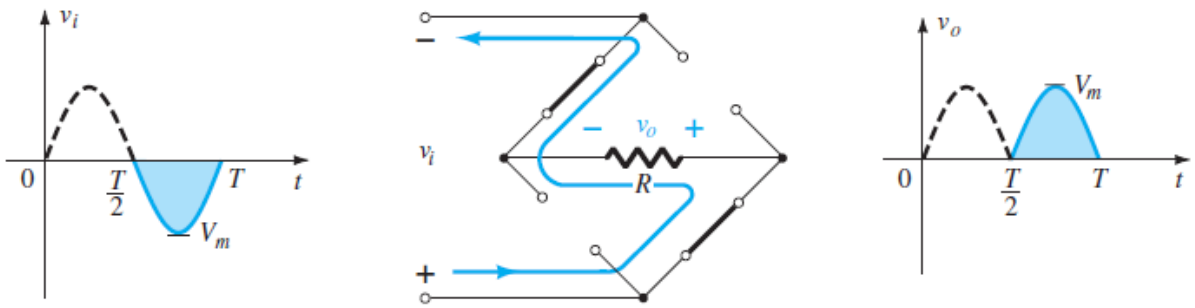

FIG. 2.56
Conduction path for the negative region of v_i.

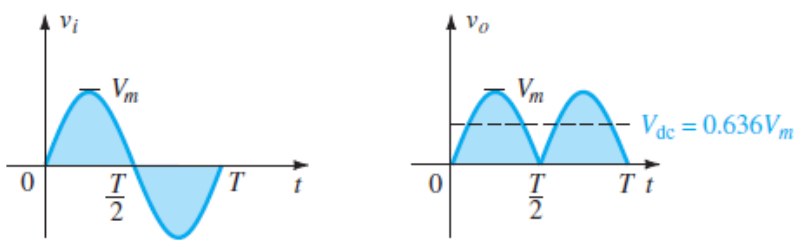

FIG. 2.57
Input and output waveforms for a full-wave rectifier.

Since the area above the axis for one full cycle is now twice that obtained for a half-wave system, the dc level has also been doubled and

$$V_{dc} = 2[\text{Eq. (2.7)}] = 2(0.318V_m)$$

or

$$\boxed{V_{dc} = 0.636\,V_m} \quad \text{full-wave} \tag{2.10}$$

If silicon rather than ideal diodes are employed as shown in Fig. 2.58, the application of Kirchhoff's voltage law around the conduction path results in

$$v_i - V_K - v_o - V_K = 0$$

and

$$v_o = v_i - 2V_K$$

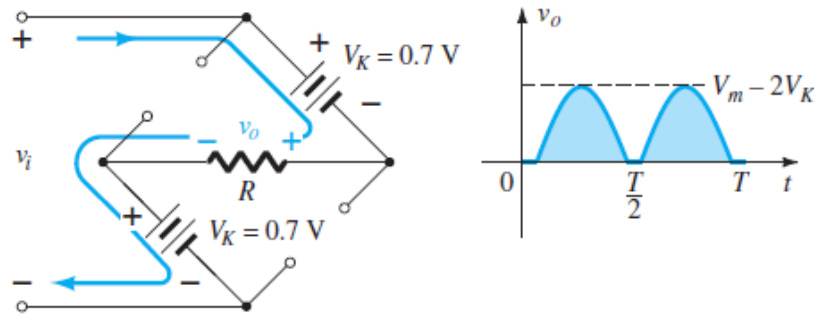

FIG. 2.58
Determining $V_{o_{max}}$ for silicon diodes in the bridge configuration.

The peak value of the output voltage v_o is therefore

$$V_{o_{max}} = V_m - 2V_K$$

For situations where $V_m \gg 2V_K$, the following equation can be applied for the average value with a relatively high level of accuracy:

$$\boxed{V_{dc} \cong 0.636(V_m - 2V_K)} \tag{2.11}$$

Then again, if V_m is sufficiently greater than $2V_K$, then Eq. (2.10) is often applied as a first approximation for V_{dc}.

PIV The required PIV of each diode (ideal) can be determined from Fig. 2.59 obtained at the peak of the positive region of the input signal. For the indicated loop the maximum voltage across R is V_m and the PIV rating is defined by

$$\boxed{\text{PIV} \geqq V_m} \quad \text{full-wave bridge rectifier} \tag{2.12}$$

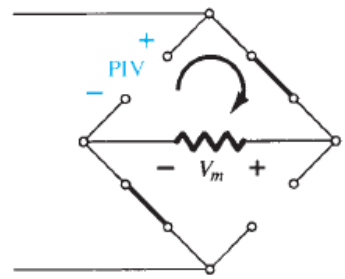

FIG. 2.59
Determining the required PIV for the bridge configuration.

Center-Tapped Transformer

A second popular full-wave rectifier appears in Fig. 2.60 with only two diodes but requiring a center-tapped (CT) transformer to establish the input signal across each section of the secondary of the transformer. During the positive portion of v_i applied to the primary of the transformer, the network will appear as shown in Fig. 2.61 with a positive pulse across each section of the secondary coil. D_1 assumes the short-circuit equivalent and D_2 the open-circuit equivalent, as determined by the secondary voltages and the resulting current directions. The output voltage appears as shown in Fig. 2.61.

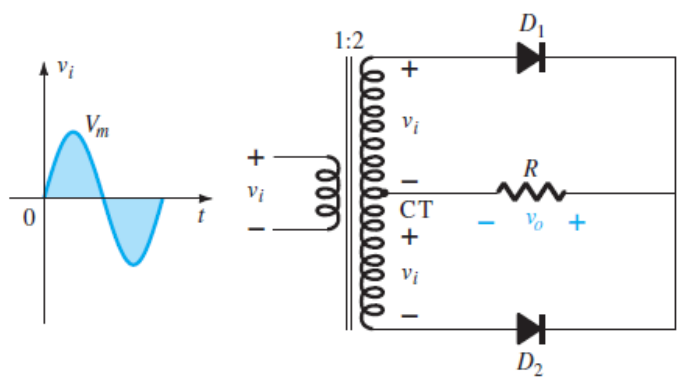

FIG. 2.60
Center-tapped transformer full-wave rectifier.

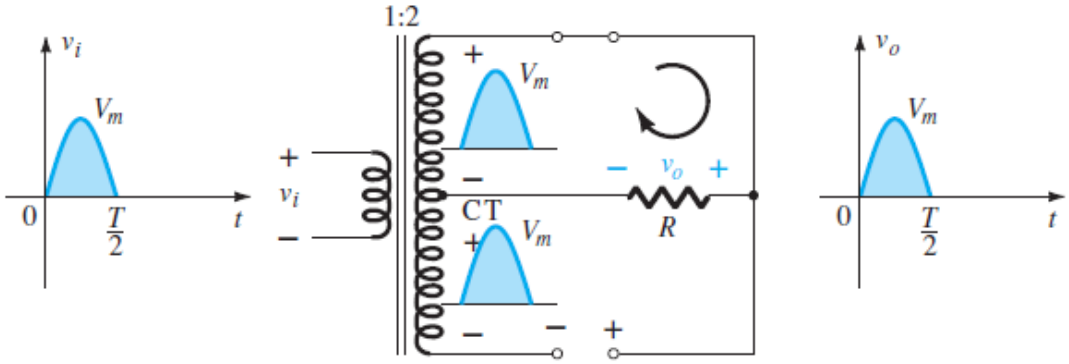

FIG. 2.61
Network conditions for the positive region of v_i.

During the negative portion of the input the network appears as shown in Fig. 2.62, reversing the roles of the diodes but maintaining the same polarity for the voltage across the load resistor R. The net effect is the same output as that appearing in Fig. 2.57 with the same dc levels.

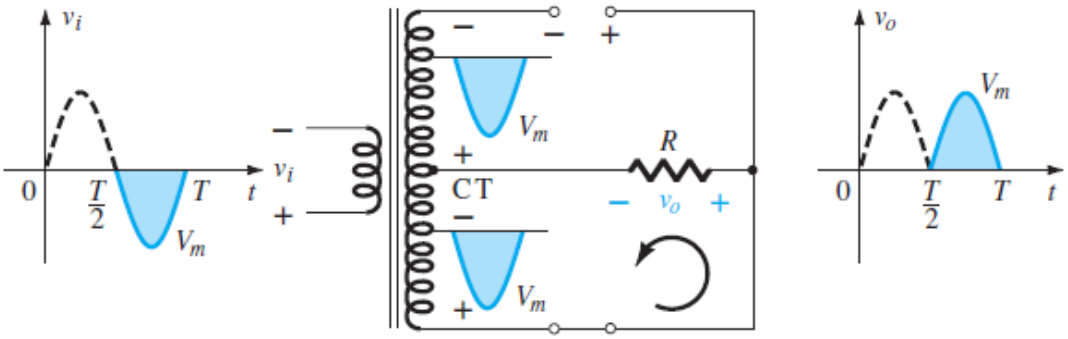

FIG. 2.62
Network conditions for the negative region of v_i.

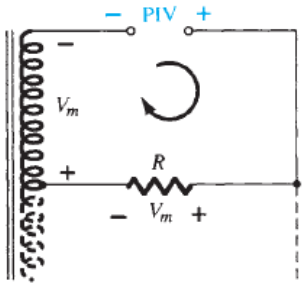

FIG. 2.63
Determining the PIV level for the diodes of the CT transformer full-wave rectifier.

PIV The network of Fig. 2.63 will help us determine the net PIV for each diode for this full-wave rectifier. Inserting the maximum voltage for the secondary voltage and V_m as established by the adjoining loop results in

$$\text{PIV} = V_{\text{secondary}} + V_R$$
$$= V_m + V_m$$

and

$$\boxed{\text{PIV} \geqq 2V_m} \quad \text{CT transformer, full-wave rectifier} \tag{2.13}$$

EXAMPLE 2.17 Determine the output waveform for the network of Fig. 2.64 and calculate the output dc level and the required PIV of each diode.

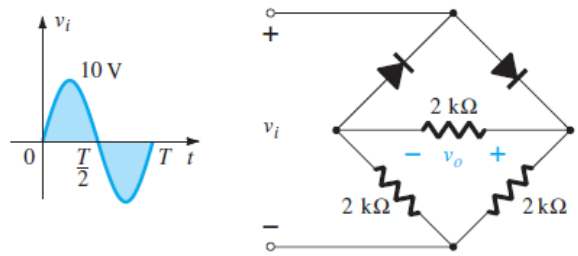

FIG. 2.64
Bridge network for Example 2.17.

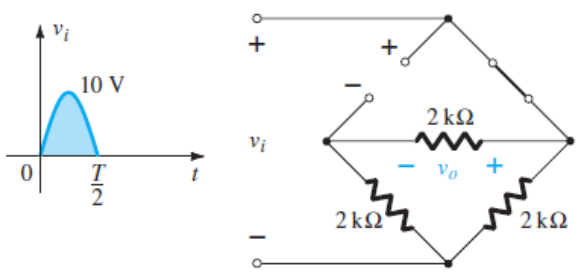

FIG. 2.65
Network of Fig. 2.64 for the positive region of v_i.

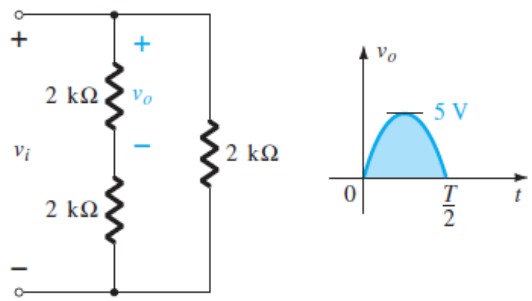

FIG. 2.66
Redrawn network of Fig. 2.65.

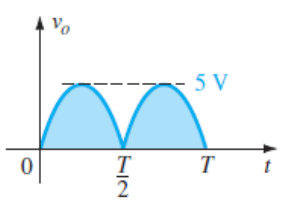

FIG. 2.67
Resulting output for Example 2.17.

Solution: The network appears as shown in Fig. 2.65 for the positive region of the input voltage. Redrawing the network results in the configuration of Fig. 2.66, where $v_o = \frac{1}{2}v_i$ or $V_{o_{\max}} = \frac{1}{2}V_{i_{\max}} = \frac{1}{2}(10\text{ V}) = 5\text{ V}$, as shown in Fig. 2.66. For the negative part of the input, the roles of the diodes are interchanged and v_o appears as shown in Fig. 2.67.

The effect of removing two diodes from the bridge configuration is therefore to reduce the available dc level to the following:

$$V_{\text{dc}} = 0.636(5\text{ V}) = \mathbf{3.18\ V}$$

or that available from a half-wave rectifier with the same input. However, the PIV as determined from Fig. 2.59 is equal to the maximum voltage across R, which is 5 V, or half of that required for a half-wave rectifier with the same input.